06/03/26

→ Signal → a physical entity that can possibly be mathematically described
↳ Image, EEC, Radio

→ Analog signal → natural
↳ not digitally transformed
↳ continuous
↳ have a resolution problem, given measurements that are too precise for our equipment
↳ within a time span

→ Digital Signal → transformed from analog to the computer (digitally)
↳ we can input the whole analog signal directly to the computer, using a small portion tends to the ∞
↳ must discretize
↳ signal is sampled at S_r sample per second
↳ discretized both in time and data itself
↳ the sampling is about time
↳ quantization is the precision between sample. The representation in the graph
↳ is digital because has sampling + Quantization

↳ only sampling is called a discrete signal

→ Time domain / Space → how something varies over time

→ Frequency domain → which frequencies are there

↳ Fourier → transform time domain to frequency domain (F^{-1} as well)

→ Fourier → bring modulus and phase → discrete-time signal (discrete Fourier)

→ DFT → to bring back to analog we must cut some parts to have

↳ number of samples must follow Nyquist's criterion

→ Nyquist's criterion → how many samples we need per second to have a digital signal that is faithful

↳ at least the double of the natural one (the highest frequency we want)

↳ we must fix a frequency ideal